

We know that if we project (transform) the data of a class using a projection vector $\mathbf{u} \in \mathbb{R}^p$ to a p dimensional subspace ($p \leq d$), i.e.:

$$\mathbf{x} \mapsto \mathbf{u}^\top \mathbf{x}, \quad (46)$$

for all data instances of the class, the mean and the covariance matrix of the class are transformed as:

$$\boldsymbol{\mu} \mapsto \mathbf{u}^\top \boldsymbol{\mu}, \quad (47)$$

$$\boldsymbol{\Sigma} \mapsto \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}, \quad (48)$$

because of characteristics of mean and variance.

The Fisher criterion (Xu & Lu, 2006) is the ratio of the between-class variance, σ_b^2 , and within-class variance, σ_w^2 :

$$f := \frac{\sigma_b^2}{\sigma_w^2} = \frac{(\mathbf{u}^\top \boldsymbol{\mu}_2 - \mathbf{u}^\top \boldsymbol{\mu}_1)^2}{\mathbf{u}^\top \boldsymbol{\Sigma}_2 \mathbf{u} + \mathbf{u}^\top \boldsymbol{\Sigma}_1 \mathbf{u}} = \frac{(\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2}{\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}}. \quad (49)$$

The FDA maximizes the Fisher criterion:

$$\underset{\mathbf{u}}{\text{maximize}} \quad \frac{(\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2}{\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}}, \quad (50)$$

which can be restated as:

$$\begin{aligned} &\underset{\mathbf{u}}{\text{maximize}} \quad (\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2, \\ &\text{subject to} \quad \mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} = 1, \end{aligned} \quad (51)$$

according to Rayleigh-Ritz quotient method (Croot, 2005). The Lagrangian (Boyd & Vandenberghe, 2004) is:

$$\mathcal{L} = (\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2 - \lambda (\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} - 1),$$

where λ is the Lagrange multiplier. Equating the derivative of $\mathcal{L}$ to zero gives:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{u}} &= 2(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \mathbf{u} - 2\lambda (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} \stackrel{\text{set}}{=} \mathbf{0} \\ \implies (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \mathbf{u} &= \lambda (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}, \end{aligned}$$

which is a generalized eigenvalue problem $((\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2, (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1))$ according to (Ghojogh et al., 2019b). The projection vector is the eigenvector of $(\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1)^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2$; therefore, we can say:

$$\mathbf{u} \propto (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1)^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2.$$

In LDA, the equality of covariance matrices is assumed. Thus, according to Eq. (18), we can say:

$$\mathbf{u} \propto (2\boldsymbol{\Sigma})^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \propto \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2. \quad (52)$$

According to Eq. (46), we have:

$$\mathbf{u}^\top \mathbf{x} \propto (\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2)^\top \mathbf{x}. \quad (53)$$

Comparing Eq. (53) with Eq. (23) shows that LDA and FDA are equivalent up to a scaling factor $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ (note that this term is multiplied as an exponential factor before taking logarithm to obtain Eq. (23), so this term a scaling factor). Hence, we can say:

$$\text{LDA} \equiv \text{FDA}. \quad (54)$$

In other words, FDA projects into a subspace. On the other hand, according to Section 7, LDA can be seen as a metric learning with a subspace where the Euclidean distance is used after projecting onto that subspace. *The two subspaces of FDA and LDA are the same subspace.* It should be noted that in manifold (subspace) learning, the scale does not matter because all the distances scale similarly.

Note that LDA assumes *one* (and not several) Gaussian for every class and so does the FDA. That is why FDA faces problem for multi-modal data (Sugiyama, 2007).

9. Relation to Logistic Regression

According to Eqs. (16) and (32), Gaussian and Bernoulli distributions are used for likelihood (class conditional) and prior, respectively, in LDA and QDA. Thus, we are making assumptions for the likelihood and prior, although we finally work with posterior in LDA and QDA according to Eq. (15). *Logistic regression* (Kleinbaum et al., 2002) says why do we make assumptions on the likelihood and prior when we want to work on posterior finally. Let us make assumption directly for the posterior.

In logistic regression, first a linear function is applied to the data to have $\beta^\top \mathbf{x}'$ where $\mathbb{R}^{d+1} \ni \mathbf{x}' = [\mathbf{x}^\top, 1]^\top$ and $\beta \in \mathbb{R}^{d+1}$ include the intercept. Then, logistic function is used in order to have a value in range $(0, 1)$ to simulate probability. Therefore, in logistic regression, the posterior is assumed to be:

$$\begin{aligned} \mathbb{P}(\mathcal{C}(\mathbf{x}) | X = \mathbf{x}) \\ = \left(\frac{\exp(\beta^\top \mathbf{x}')}{1 + \exp(\beta^\top \mathbf{x}')} \right)^{\mathcal{C}(\mathbf{x})} \left(\frac{1}{1 + \exp(\beta^\top \mathbf{x}')} \right)^{1 - \mathcal{C}(\mathbf{x})}, \end{aligned} \quad (55)$$

where $\mathcal{C}(\mathbf{x}) \in \{-1, +1\}$ for the two classes. Logistic regression considers the coefficient β as the parameter to be optimized and uses Newton's method (Boyd & Vandenberghe, 2004) for the optimization. Therefore, in summary, logistic regression makes assumption on the posterior while LDA and QDA make assumption on likelihood and prior.

10. Relation to Bayes Optimal Classifier and Gaussian Naive Bayes

The Bayes classifier maximizes the posteriors of the classes (Murphy, 2012):

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \mathbb{P}(\mathbf{x} \in \mathcal{C}_k | X = \mathbf{x}). \quad (56)$$

According to Eq. (14) and Bayes rule, we have:

$$\mathbb{P}(\mathbf{x} \in \mathcal{C}_k | X = \mathbf{x}) \propto \mathbb{P}(X = \mathbf{x} | \mathbf{x} \in \mathcal{C}_k) \underbrace{\mathbb{P}(\mathbf{x} \in \mathcal{C}_k)}_{\pi_k}, \quad (57)$$

where the denominator of posterior (the marginal) which is:

$$\mathbb{P}(X = \mathbf{x}) = \sum_{r=1}^{|\mathcal{C}|} \mathbb{P}(X = \mathbf{x} | \mathbf{x} \in \mathcal{C}_r) \pi_r,$$

is ignored because it is not dependent on the classes $\mathcal{C}_1$ to $\mathcal{C}_{|\mathcal{C}|}$.

According to Eq. (57), the posterior can be written in terms of likelihood and prior; therefore, Eq. (56) can be restated as:

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \pi_k \mathbb{P}(X = \mathbf{x} | \mathbf{x} \in \mathcal{C}_k). \quad (58)$$

Note that the Bayes classifier does not make any assumption on the posterior, prior, and likelihood, unlike LDA and QDA which assume the uni-modal Gaussian distribution for the likelihood (and we *may* assume Bernoulli distribution for the prior in LDA and QDA according to Eq. (32)). Therefore, we can say the difference of Bayes and QDA is in assumption of *uni-modal* Gaussian distribution for the likelihood (class conditional); hence, if the likelihoods are already uni-modal Gaussian, the Bayes classifier reduces to QDA. Likewise, the difference of Bayes and LDA is in assumption of Gaussian distribution for the likelihood (class conditional) and equality of covariance matrices of classes; thus, if the likelihoods are already Gaussian and the covariance matrices are already equal, the Bayes classifier reduces to LDA.

It is noteworthy that the Bayes classifier is an optimal classifier because it can be seen as an ensemble of hypotheses (models) in the hypothesis (model) space and no other ensemble of hypotheses can outperform it (see Chapter 6, Page 175 in (Mitchell, 1997)). In the literature, it is referred to as *Bayes optimal classifier*. To better formulate the explained statements, the Bayes optimal classifier estimates the class as:

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_{\mathcal{C}_k \in \mathcal{C}} \sum_{h_j \in \mathcal{H}} \mathbb{P}(\mathcal{C}_k | h_j) \mathbb{P}(\mathcal{D} | h_j) \mathbb{P}(h_j), \quad (59)$$

where $\mathcal{C} := \{\mathcal{C}_1, \dots, \mathcal{C}_{|\mathcal{C}|}\}$, $\mathcal{D} := \{\mathbf{x}_i\}_{i=1}^n$ is the training set, h_j is a hypothesis for estimating the class of instances, and $\mathcal{H}$ is the hypothesis space including all possible hypotheses.

According to Bayes rule, similar to what we had for Eq. (57), we have:

$$\mathbb{P}(h_j | \mathcal{D}) \propto \mathbb{P}(\mathcal{D} | h_j) \mathbb{P}(h_j).$$

Therefore, Eq. (59) becomes (Mitchell, 1997):

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_{\mathcal{C}_k \in \mathcal{C}} \sum_{h_j \in \mathcal{H}} \mathbb{P}(\mathcal{C}_k | h_j) \mathbb{P}(h_j | \mathcal{D}), \quad (60)$$

In conclusion, the Bayes classifier is optimal. Therefore, if the likelihoods of classes are Gaussian, QDA is an optimal classifier and if the likelihoods are Gaussian and the covariance matrices are equal, the LDA is an optimal classifier. Often, the distributions in the natural life are Gaussian; especially, because of central limit theorem (Hazewinkel, 2001), the summation of independent and identically distributed (iid) variables is Gaussian and the signals usually add in the real world. This explains why LDA and QDA are very effective classifiers in machine learning. We also saw that FDA is equivalent to LDA. Thus, the reason of effectiveness of the powerful FDA classifier becomes clear. We have seen the very successful performance of FDA and LDA in different applications, such as face recognition (Belhumeur et al., 1997; Etemad & Chellappa, 1997; Zhao et al., 1999), action recognition (Ghojogh et al., 2017; Mokari et al., 2018), and EEG classification (Malek Mohammadi et al., 2019).

Implementing Bayes classifier is difficult in practice so we approximate it by *naive Bayes* (Zhang, 2004). If x_j denotes the j -th dimension (feature) of $\mathbf{x} = [x_1, \dots, x_d]^T$, Eq. (58) is restated as:

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \pi_k \mathbb{P}(x_1, x_2, \dots, x_d | \mathbf{x} \in \mathcal{C}_k). \quad (61)$$

The term $\mathbb{P}(x_1, x_2, \dots, x_d | \mathbf{x} \in \mathcal{C}_k)$ is very difficult to compute as the features are possibly correlated. Naive Bayes relaxes this possibility and naively assumes that the features are conditionally independent ($\perp\!\!\!\perp$) when they are conditioned on the class:

$$\begin{aligned} \mathbb{P}(x_1, x_2, \dots, x_d | \mathbf{x} \in \mathcal{C}_k) \\ \stackrel{\perp\!\!\!\perp}{\approx} \mathbb{P}(x_1 | \mathcal{C}_k) \mathbb{P}(x_2 | \mathcal{C}_k) \cdots \mathbb{P}(x_d | \mathcal{C}_k) = \prod_{j=1}^d \mathbb{P}(x_j | \mathcal{C}_k). \end{aligned}$$

Therefore, Eq. (61) becomes:

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \pi_k \prod_{j=1}^d \mathbb{P}(x_j | \mathcal{C}_k). \quad (62)$$

In *Gaussian naive Bayes*, univariate Gaussian distribution is assumed for the likelihood (class conditional) of every feature:

$$\mathbb{P}(x_j | \mathcal{C}_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x_j - \mu_k)^2}{2\sigma_k^2}\right), \quad (63)$$

where the mean and unbiased variance are estimated as:

$$\mathbb{R} \ni \hat{\mu}_k = \frac{1}{n_k} \sum_{i=1}^n x_{i,j} \mathbb{I}(\mathcal{C}(x_i) = k), \quad (64)$$

$$\mathbb{R} \ni \hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{i=1}^n (x_{i,j} - \hat{\mu}_k)^2 \mathbb{I}(\mathcal{C}(x_i) = k), \quad (65)$$

where $x_{i,j}$ denotes the j -th feature of the i -th training instance. The prior can again be estimated using Eq. (32).

According to Eqs. (62) and (63), Gaussian naive Bayes is equivalent to QDA where the covariance matrices are *diagonal*, i.e., the off-diagonal of the covariance matrices are ignored. Therefore, we can say that QDA is more powerful than Gaussian naive Bayes because Gaussian naive Bayes is a simplified version of QDA. Moreover, it is obvious that Gaussian naive Bayes and QDA are equivalent for one dimensional data. Comparing to LDA, the Gaussian naive Bayes is equivalent to LDA if the covariance matrices are diagonal and they are all equal, i.e., $\sigma_1^2 = \dots = \sigma_{|C|}^2$; therefore, LDA and Gaussian naive Bayes have their own assumptions, one on the off-diagonal of covariance matrices and the other one on equality of the covariance matrices. As Gaussian naive Bayes has some level of optimality (Zhang, 2004), it becomes clear why LDA and QDA are such effective classifiers.

11. Relation to Likelihood Ratio Test

Consider two hypotheses for estimating some parameter, a null hypothesis H_0 and an alternative hypothesis H_A . The probability $\mathbb{P}(\text{reject } H_0 | H_0)$ is called type 1 error, false positive error, or false alarm error. The probability $\mathbb{P}(\text{accept } H_0 | H_A)$ is called type 2 error or false negative error. The $\mathbb{P}(\text{reject } H_0 | H_0)$ is also called *significance level*, while $1 - \mathbb{P}(\text{accept } H_0 | H_A) = \mathbb{P}(\text{reject } H_0 | H_A)$ is called *power*.

If $L(\theta_A)$ and $L(\theta_0)$ are the likelihoods (probabilities) for the alternative and null hypotheses, the likelihood ratio is:

$$\Lambda = \frac{L(\theta_A)}{L(\theta_0)} = \frac{f(\mathbf{x}; \theta_A)}{f(\mathbf{x}; \theta_0)}. \quad (66)$$

The Likelihood Ratio Test (LRT) (Casella & Berger, 2002) rejects the H_0 in favor of H_A if the likelihood ratio is greater than a threshold, i.e., $\Lambda \geq t$. The LRT is a very effective statistical test because according to the Neyman-Pearson lemma (Neyman & Pearson, 1933), it has the largest power among all statistical tests with the same significance level.

If the sample size is large, $n \rightarrow \infty$, and the θ_A is estimated using MLE, the logarithm of the likelihood ratio asymptotically has the distribution of χ^2 under the null hypothesis (White, 1984; Casella & Berger, 2002):

$$2 \ln(\Lambda) \overset{H_0}{\sim} \chi_{(df)}^2, \quad (67)$$

where the degree of freedom of χ^2 distribution is $df := \dim(H_A) - \dim(H_0)$ and $\dim(\cdot)$ is the number of unspecified parameters in the hypothesis.

There is a connection between LDA or QDA and the LRT (Lachenbruch & Goldstein, 1979). Recall Eq. (12) or (15) which can be restated as:

$$\frac{f_2(\mathbf{x}) \pi_2}{f_1(\mathbf{x}) \pi_1} = 1, \quad (68)$$

which is for the decision boundary. The Eq. (22) dealt with the difference of $f_2(x) \pi_2$ and $f_1(x) \pi_1$; however, here we are dealing with their ratio. Recall Fig. 1 where if we move x^* to the right and left, the ratio $f_2(x^*) \pi_2 / f_1(x^*) \pi_1$ decreases and increases, respectively, because the probabilities of the first and second class happening change. In other words, moving the x^* changes the significance level and power. Therefore, Eq. (68) can be used to have a statistical test where the posteriors are used in the ratio, as we also used posteriors in LDA and QDA. The null/alternative hypothesis can be considered to be the mean and covariance of the first/second class. In other words, the two hypotheses say that the point belongs to a specific class. Hence, if the ratio is larger than a value t , the instance $\mathbf{x}$ is estimated to belong to the second class; otherwise, the first class is chosen. According to Eq. (16), the Eq. (68) becomes:

$$\frac{(|\Sigma_2|)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^\top \Sigma_2^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\right) \pi_2}{(|\Sigma_1|)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^\top \Sigma_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \pi_1} \geq t, \quad (69)$$

for QDA. In LDA, the covariance matrices are equal, so:

$$\frac{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_2)^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_2)\right) \pi_2}{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \pi_1} \geq t. \quad (70)$$

As can be seen, changing the priors change impacts the ratio as expected. Moreover, the value of t can be chosen according to the desired significance level in the χ^2 distribution using the χ^2 table. The Eqs. (69) and (70) show the relation of LDA and QDA with LRT. As the LRT has the largest power (Neyman & Pearson, 1933), the effectiveness of LDA and QDA in classification is explained from a hypothesis testing point of view.

12. Simulations

In this section, we report some simulations which make the concepts of tutorial clearer by illustration.

12.1. Experiments with Equal Class Sample Sizes

We created a synthetic dataset of three classes each of which is a two dimensional Gaussian distribution. The means and covariance matrices of the three Gaussians from